Algorithmic Geometry Fall, 2026

TD 4

1. Give a lower bound that shows that with $c = 1$ may construct a coloring with discrepancy $\Omega(n)$.

Solution. We use an adversary argument. We fix $k \in \mathbb{N}$, and we set $n = 2k + 1$. Our set system consists of just three sets A_1, A_2, and B. We assign the vertices $v_1, \ldots, v_n$ to the sets, in an online manner.

On the arrival of v_1: add v_1 into A_1 and A_2. The sum discrepancy algorithm sets $\chi(v_1) = +1$, since all sets containing v_1 have discrepancy 0.

On the arrival of v_2: add v_2 into A_1, A_2, and B. The sum discrepancy algorithm sets $\chi(v_2) = -1$, since this yields a total discrepancy of 1 (otherwise, we would get total discrepancy 5).

On the arrival of v_3: add v_3 into A_1 and A_2. The sum discrepancy algorithm sets $\chi(v_3) = +1$, (both choices for $\chi(v_3)$ yield total discrepancy 3, so we may assume that the sum discrepancy algorithm uses the color $+1$).

On the arrival of v_4: add v_4 into A_1, A_2, and B. The sum discrepancy algorithm sets $\chi(v_4) = -1$, since this yields a total discrepancy of 2 (otherwise, we would get total discrepancy 4).

Situation before iteration $2t + 1$: We have $\operatorname{disc}^{2t+1}(A_1) = \operatorname{disc}^{2t+1}(A_2) = 0$, and $\operatorname{disc}^{2t+1}(B) = t$.

On the arrival of v_{2t+1}: add v_{2t+1} into A_1 and A_2. The sum discrepancy algorithm sets $\chi(v_{2t+1}) = +1$, (both choices lead to a discrepancy of $t + 2$, so that the sum discrepancy algorithm uses the color $+1$).

On the arrival of v_{2t+2}: add v_{2t+2} into A_1, A_2, and B. The sum discrepancy algorithm sets $\chi(v_{2t+2}) = -1$, since this yields a total discrepancy of $t + 1$ (otherwise, we would get total discrepancy $t + 3$.)

After iteration $2k$, the set B has discrepancy $k = \lfloor n/2 \rfloor$. In the end, we add $v_n = v_{2k+1}$ to only A_1, so that the sets A_1 and A_2 are different. This does not affect the discrepancy of B.

2. We now consider a famous problem where the greedy algorithm actually does give a good solution.

   Let $V = \{v_1, \ldots, v_n\}$ be n elements, and $\mathcal{S} = \{S_1, \ldots, S_m\}$ be m sets. Our goal is to compute a small set $H \subseteq V$ that is a hitting set for $\mathcal{S}$. Let OPT be the size of a minimum hitting set for $\mathcal{S}$.

   Prove that the following algorithm returns a hitting set H of size at most $\text{OPT} \cdot \ln m$.

> ***Set Cover*** $\left(V = \{v_1, \ldots, v_n\}, \mathcal{S} = \{S_1, \ldots, S_m\}\right)$
>
> Set $\omega^1(S) = 1$ for all $S \in \mathcal{S}$.
> Set $H = \emptyset$.
> For iteration $t = 1, \ldots, T$:
>
> (a) pick an element $v^t \in V$ with:
>
> $$v^t = \arg\max_{v \in V} \sum_{\substack{S \in \mathcal{S}: \\ v \in S}} \omega^t(S).$$
>
> (b) add v^t to H, and set the weights of all sets of $\mathcal{S}$ hit by v^t to 0:
>
> $$\omega^{t+1}(S) = \begin{cases} 0 & \text{if } v^t \in S \\ \omega^t(S) & \text{otherwise.} \end{cases}$$
>
> Return H.

Solution. Set $\Omega^t = \sum_{S \in \mathcal{S}} \omega^t(S)$.

The key idea is that at each step, by the pigeonhole principle, there is an element of the optimal hitting set that hits sets of total weight at least

$$\frac{\Omega^t}{\text{OPT}},$$

and thus we have

$$\Omega^{t+1} \leq \Omega^t - \frac{\Omega^t}{\text{OPT}} = \Omega^t \cdot \left(1 - \frac{1}{\text{OPT}}\right).$$

After T steps, assume that H is not a hitting set. Then we have

$$1 \leq \Omega^{T+1} \leq m \cdot \left(1 - \frac{1}{\text{OPT}}\right)^T.$$

Simplifying,

$$\ln 1 \leq \ln m - \frac{T}{\text{OPT}}.$$

Thus we can conclude that

$$|H| = T \leq \text{OPT} \cdot \ln m.$$